

Smytten E-Commerce Database Management System

CHAPTER – 2

Identify entities, attributes, primary keys, and relationships required for the system.

Description:

In this phase of the project, the entities, attributes, primary keys, foreign keys, and relationships required for the Smytten E-Commerce Database Management System were identified. These components form the basic structure of the database and help organize data efficiently. Identifying these elements is an important step in designing a relational database that ensures data consistency, minimizes redundancy, and supports smooth e-commerce operations.

2.1 Entities, Attributes and Primary Keys:

Customer

Attribute	Description	Key
Customer_ID	Customer ID	Primary Key
Customer_Name	Customer Name	-
Email	Email Address	-
Phone Number	Phone Number	-
Address	Customer Address	-

Admin

Attribute	Description	Key
Admin_ID	Admin ID	Primary Key
Admin Name	Admin Name	-
Email	Official Email	-
Password	Login Password	-

Brand

Attribute	Description	Key
Brand_ID	Brand ID	Primary Key
Brand Name	Brand Name	-
Brand Type	Beauty/Wellness	-

Category

Attribute	Description	Key
Category_ID	Category ID	Primary Key
Category_Name	Category Name	-

Product

Attribute	Description	Key
Product_ID	Product ID	Primary Key
Product_Name	Product Name	-
Price	Product Price	-
Stock	Available Stock	-
Brand_ID	References Brand	Foreign Key
Category_ID	References Category	Foreign Key

Cart

Attribute	Description	Key
Cart_ID	Cart ID	Primary Key
Customer_ID	References Customer	Foreign Key
Product_ID	References Product	Foreign Key
Quantity	Quantity	-

Wishlist

Attribute	Description	Key
Wishlist_ID	Wishlist ID	Primary Key
Customer_ID	References Customer	Foreign Key
Product_ID	References Product	Foreign Key

Orders

Attribute	Description	Key
Order_ID	Order ID	Primary Key
Customer_ID	References Customer	Foreign Key
Order_Date	Date of Order	-
Total_Amount	Total Amount	-
Order_Status	Pending/Shipped/Delivered	-

Payment

Attribute	Description	Key
Payment_ID	Payment ID	Primary Key
Order_ID	References Orders	Foreign Key
Payment_Mode	UPI/Card/COD	-
Payment_Status	Success/Failed	-

Delivery

Attribute	Description	Key
Delivery_ID	Delivery ID	Primary Key
Order_ID	References Orders	Foreign Key
Delivery_Partner	Courier Service	-
Delivery_Status	Shipped/Delivered	-

Review

Attribute	Description	Key
Review_ID	Review ID	Primary Key
Customer_ID	References Customer	Foreign Key
Product_ID	References Product	Foreign Key
Rating	Product Rating	-
Review_Text	Customer Feedback	-

Inventory

Attribute	Description	Key
Inventory_ID	Inventory ID	Primary Key
Product_ID	References Product	Foreign Key
Available_Stock	Current Stock	-
Last_Updated	Last Updated	-

2.2 Relationships:

Entity 1	Relationship	Entity 2
Customer	1 : M	Orders
Customer	1 : M	Cart
Customer	1 : M	Wishlist
Brand	1 : M	Product
Category	1 : M	Product
Orders	1 : 1	Payment
Orders	1 : 1	Delivery
Product	1 : M	Review
Product	1 : 1	Inventory